Experiment 36: Stock Price Variability Analysis

Aim: To calculate the variability of stock prices using standard deviation and variance.

Algorithm:

1. Import Pandas.
2. Load stock data from a CSV file.
3. Calculate mean closing price.
4. Calculate variance and standard deviation.
5. Display the results.

Code:

```
import pandas as pd

df = pd.DataFrame({
   "Close": [100, 105, 110, 95, 120, 115]
})

print("Mean:", df["Close"].mean())
print("Variance:", df["Close"].var())
print("Standard Deviation:", df["Close"].std())
```

Input:

Closing Prices: 100, 105, 110, 95, 120, 115

Output:

Mean: 107.5

Variance: 87.5

Standard Deviation: 9.35



Result: Stock price variability was calculated successfully.

Note: This example uses sample data in a DataFrame. For CSV input, use pd.read_csv("stock.csv").